

Reinforcement Learning (RL): Core Concepts

Reinforcement Learning (RL)

- The main concept on which RL works is an agent learning by trial and error.
- Unlike other types of machine learning, the agent is not explicitly told what the correct action is. It must discover which actions yield the most reward.
- Rewards may be delayed, meaning an action taken now can result in positive or negative rewards much later in the future (after multiple steps in a trajectory).

Environment & The Loop

Environment: The world that the agent lives in and interacts with.

The Loop: Agent sees a state $\rightarrow$ takes an action $\rightarrow$ receives a reward signal and the next state from the environment.

The environment can control when to ask the agent to act.

Key Terms

1. **State (s):** A complete description of the world.
2. **Observation (o):** A partial description of the state.
3. **Action Space:** The set of all valid actions available to the agent.
4. **Trajectories:** A sequence of states and actions over time.
5. **Policy (π):** Rule by which agent takes action.
6. **Return (G):** Total reward over a trajectory.
7. **Discount Factor (γ):** Factor to ensure the mathematical sum of future rewards converges.

The Elements of Reinforcement Learning

Policy

- A policy defines the learning agent's way of behaving at a given time. It is a mapping from the perceived states of the environment to the actions to be taken when in those states.

- A policy can be as simple as a basic lookup table or function, or it can involve extensive computation.
- The policy is the core of the RL agent because it alone is sufficient to determine behavior.

Return

- It is the Reward signal which defines the immediate goal of the RL problem.
- If an action chosen by the policy is followed by low reward, the policy will be updated to select a different action in that situation next time.

Value Functions

Functions that estimate the expected return to predict how good a situation is:

- **State-Value Function (V):** Measures how good it is to be in a specific state.
- **Action-Value Function (Q):** Measures how good it is to take a specific action in a specific state.
- **Optimal Value Functions (V^*, Q^*):** The maximum possible expected return an agent can achieve.

Exploration vs. Exploitation

- The fundamental trade-off the agent faces while learning.
- **Exploration:** Trying out new, untested actions to gather more information about the environment.
- **Exploitation:** Using known information to choose the action that is currently expected to yield the highest reward.
- An agent must balance both i.e. exploring too much wastes time, but exploiting too early means missing out on potentially better strategies.